

LEARNQUEST

LINUX LABS DOCUMENTATION

MODULE 3

File, Directory & Link Management

Command-Line Practice Documentation

Linux Fundamentals Coursework

July 2026

1. Introduction

The Linux command line is the foundation of effective system administration, software development, and everyday computing on Linux-based systems. Unlike graphical interfaces, the terminal gives users direct, precise control over the file system — allowing files and directories to be created, organized, inspected, searched, compared, and linked with a small set of powerful commands.

This module documents hands-on practice with core Linux file-management utilities. Rather than presenting the commands as a single, undifferentiated list, they are organized below by functional category — Directory Management, File Management, Viewing Files, Searching, Comparing Files, and Links — so that related commands can be learned, referenced, and revised together. Each category includes a summary table of the commands practiced, followed by terminal screenshots demonstrating the corresponding output.

2. Learning Objectives

By the end of this module, the learner will be able to:

- Explain the role of the Linux command-line interface in managing files and directories.
- Create and remove directories and files using `mkdir`, `touch`, and `rm`.
- List and inspect file attributes using various options of the `ls` command.
- View file contents in different ways using `cat`, `head`, and `tail`.
- Search for text within files using `grep`, including case-insensitive matching.
- Compare files and directories line by line, by summary, or side by side using `diff`.
- Create, distinguish, and manage hard links and symbolic (soft) links using `ln`.

3. Learned Path

The commands in this module build on one another in a natural, practical sequence:

Step 1 — Set up the workspace: Create a directory to work in using `mkdir`, then populate it with files using `touch`.

Step 2 — Manage files: Practice removing files and directories with `rm` to keep the workspace clean.

Step 3 — View and inspect: Use the many options of `ls`, along with `cat`, `head`, and `tail`, to list files and read their contents.

Step 4 — Search: Use `grep` to locate specific text inside files quickly and accurately.

Step 5 — Compare: Use `diff` to identify differences between files and directories.

Step 6 — Link: Use `ln` to create hard and symbolic links, and observe how each behaves when the original file is modified or removed.

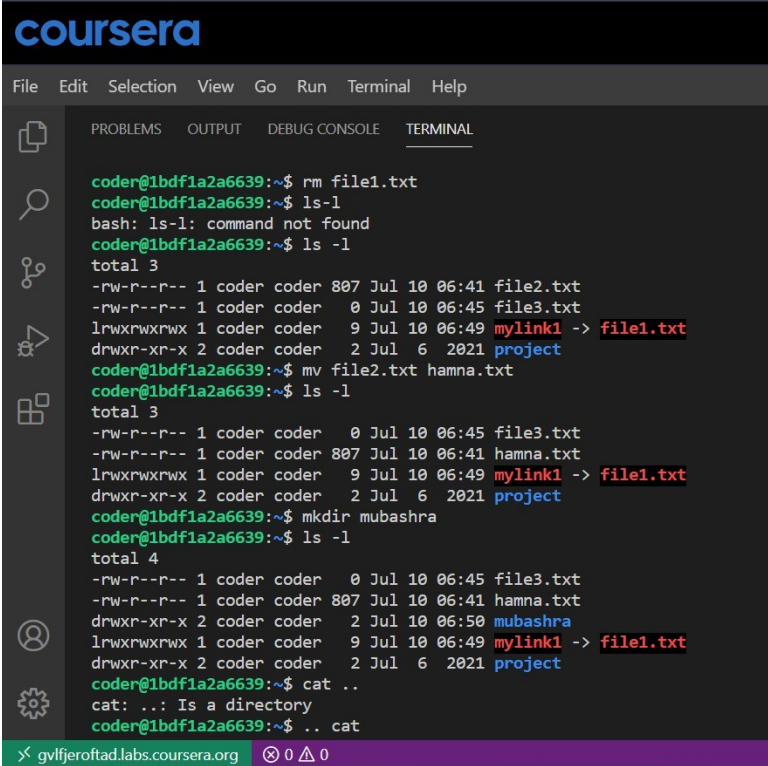
4. Directory Management

Command	Description
<code>mkdir</code>	Creates a new, empty directory at the specified location.

No dedicated terminal screenshot for `mkdir` was captured in this module; its effect is visible as the pre-existing "project" directory referenced in the screenshots below.

5. File Management

Command	Description
touch	Creates a new, empty file in the current directory (or updates its timestamp if it already exists).
rm	Deletes (removes) the specified file or directory from the file system.
rm -r	Recursively removes a directory and all of its contents.

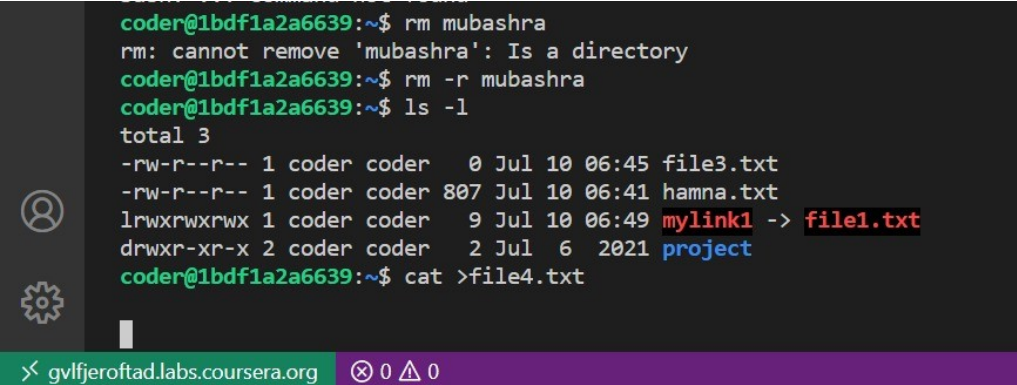


```
coursera
File Edit Selection View Go Run Terminal Help

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

coder@1bdf1a2a6639:~$ rm file1.txt
coder@1bdf1a2a6639:~$ ls -l
bash: ls-l: command not found
coder@1bdf1a2a6639:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ mv file2.txt hamna.txt
coder@1bdf1a2a6639:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 hamna.txt
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ mkdir mubashra
coder@1bdf1a2a6639:~$ ls -l
total 4
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 hamna.txt
drwxr-xr-x 2 coder coder 2 Jul 10 06:50 mubashra
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ cat ..
cat: ..: Is a directory
coder@1bdf1a2a6639:~$ .. cat
gvlfjeroftad.labs.coursera.org 0 0 0
```

Figure 1 — Removing a file with `rm` and confirming with `ls -l`.

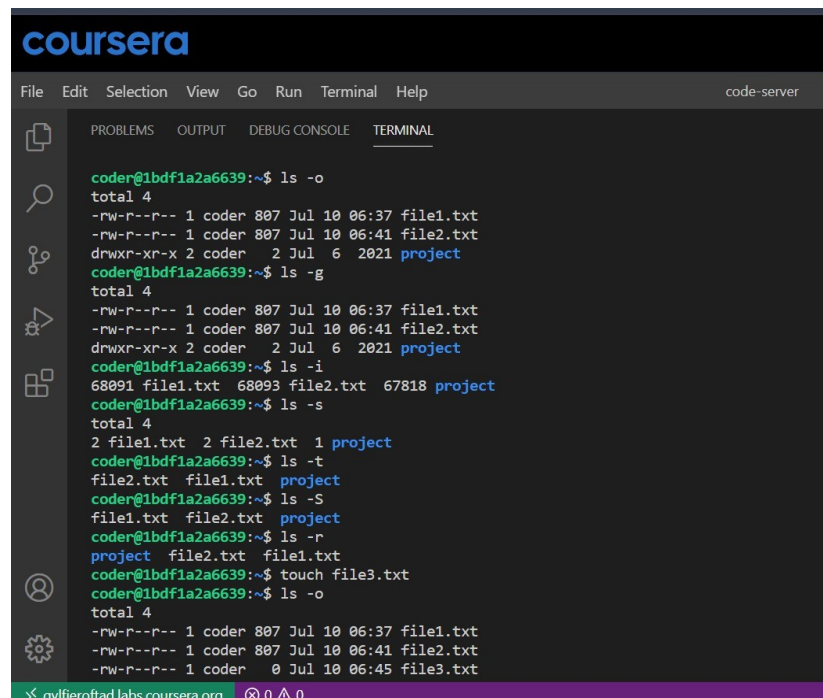


```
coder@1bdf1a2a6639:~$ rm mubashra
rm: cannot remove 'mubashra': Is a directory
coder@1bdf1a2a6639:~$ rm -r mubashra
coder@1bdf1a2a6639:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 hamna.txt
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ cat >file4.txt
gvlfjeroftad.labs.coursera.org 0 0 0
```

Figure 2 — Recursively removing a directory (`rm -r`) and creating a file with `cat > file`.

6. Viewing Files

Command	Description
<code>ls -o</code>	Lists files in long format without displaying the group name.
<code>ls -g</code>	Lists files in long format without displaying the owner name.
<code>ls -i</code>	Lists files along with their inode (index) numbers.
<code>ls -s</code>	Lists files together with their allocated size.
<code>ls -lt</code>	Lists files sorted by modification time, most recent first.
<code>ls -lS</code>	Lists files sorted by size, largest first.
<code>ls -lr</code>	Lists files in reverse order.
<code>cat > file</code>	Creates a new file and lets you type its contents directly.
<code>cat f1 f2 > f3</code>	Combines (concatenates) the contents of two files into a third file.
<code>cat -n file</code>	Displays file contents with line numbers.
<code>cat -s file</code>	Suppresses repeated blank lines in the output.
<code>head -1 file</code>	Displays the first N lines of a file (here, the first line).
<code>tail -1 file</code>	Displays the last N lines of a file (here, the last line).



```
coursera
File Edit Selection View Go Run Terminal Help code-server

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

coder@1bdf1a2a6639:~$ ls -o
total 4
-rw-r--r-- 1 coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder 807 Jul 10 06:41 file2.txt
drwxr-xr-x 2 coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ls -g
total 4
-rw-r--r-- 1 coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder 807 Jul 10 06:41 file2.txt
drwxr-xr-x 2 coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ls -i
68091 file1.txt 68093 file2.txt 67818 project
coder@1bdf1a2a6639:~$ ls -s
total 4
2 file1.txt 2 file2.txt 1 project
coder@1bdf1a2a6639:~$ ls -t
file2.txt file1.txt project
coder@1bdf1a2a6639:~$ ls -S
file1.txt file2.txt project
coder@1bdf1a2a6639:~$ ls -r
project file2.txt file1.txt
coder@1bdf1a2a6639:~$ touch file3.txt
coder@1bdf1a2a6639:~$ ls -o
total 4
-rw-r--r-- 1 coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder 0 Jul 10 06:45 file3.txt
```

Figure 3 — `ls -o` listing files without the group column.

```

drwxr-xr-x 2 coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ls -lt
total 4
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ls -ls
total 4
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
coder@1bdf1a2a6639:~$ ls -lr
total 4
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
coder@1bdf1a2a6639:~$ ln file3.txt mylink
coder@1bdf1a2a6639:~$ ls
file1.txt file2.txt file3.txt mylink project
coder@1bdf1a2a6639:~$ ls -o
total 5
-rw-r--r-- 1 coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 2 coder 0 Jul 10 06:45 file3.txt

```

Figure 4 — `ls -lt` listing files sorted by modification time.

```

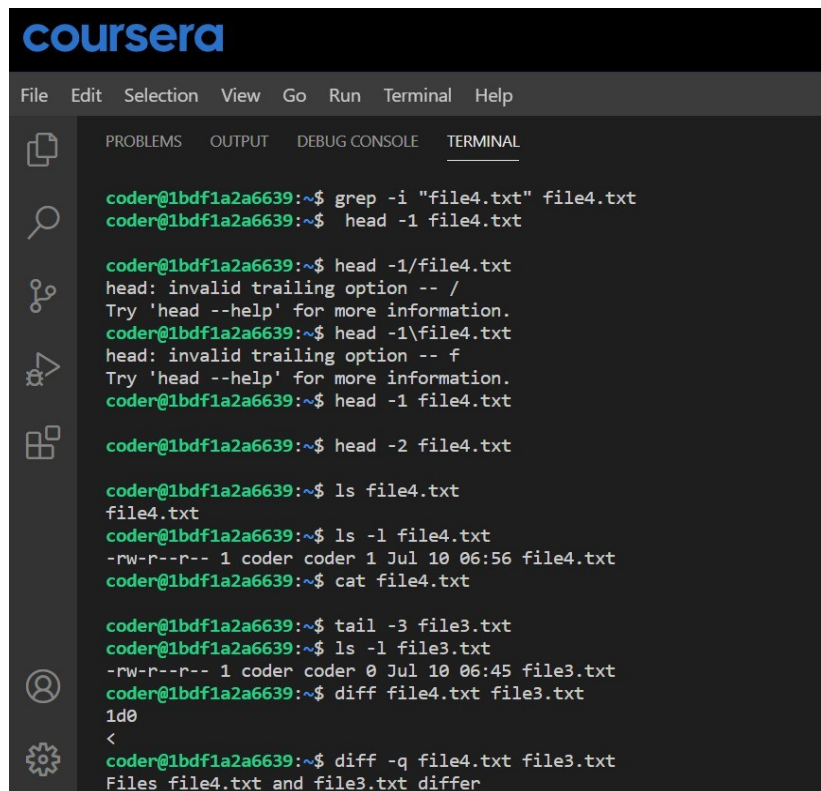
project:
coder@1bdf1a2a6639:~$ ls -l mubashra
total 0
coder@1bdf1a2a6639:~$ ls -l project
total 0
coder@1bdf1a2a6639:~$ ls -l file3.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
coder@1bdf1a2a6639:~$ ls -l file4.txt
-rw-r--r-- 1 coder coder 1 Jul 10 06:56 file4.txt
coder@1bdf1a2a6639:~$ diff -y <(ls -l file3.txt) <(ls -l file4.txt)
bash: syntax error near unexpected token `('
coder@1bdf1a2a6639:~$ diff -y <(ls -l file3.txt) <(ls -l file4.txt)
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt          | -rw-r--r-- 1 coder coder 1 Jul 10 06:56 file4.txt
coder@1bdf1a2a6639:~$ which diff file3.txt
/usr/bin/diff
coder@1bdf1a2a6639:~$ locate mubashra
coder@1bdf1a2a6639:~$ locate file3.txt
coder@1bdf1a2a6639:~$ which locate
/usr/bin/locate
coder@1bdf1a2a6639:~$ locate file3.txt
coder@1bdf1a2a6639:~$ which locate file4.txt
/usr/bin/locate
coder@1bdf1a2a6639:~$

```

Figure 5 — `ls -l` used on individual files and directories.

7. Searching

Command	Description
<code>grep -i "str" file</code>	Searches for a string inside a file; -i makes the match case-insensitive.



```
coursera
File Edit Selection View Go Run Terminal Help

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

coder@1bdf1a2a6639:~$ grep -i "file4.txt" file4.txt
coder@1bdf1a2a6639:~$ head -1 file4.txt

coder@1bdf1a2a6639:~$ head -1/file4.txt
head: invalid trailing option -- /
Try 'head --help' for more information.
coder@1bdf1a2a6639:~$ head -1\file4.txt
head: invalid trailing option -- f
Try 'head --help' for more information.
coder@1bdf1a2a6639:~$ head -1 file4.txt

coder@1bdf1a2a6639:~$ head -2 file4.txt

coder@1bdf1a2a6639:~$ ls file4.txt
file4.txt
coder@1bdf1a2a6639:~$ ls -l file4.txt
-rw-r--r-- 1 coder coder 1 Jul 10 06:56 file4.txt
coder@1bdf1a2a6639:~$ cat file4.txt

coder@1bdf1a2a6639:~$ tail -3 file3.txt
coder@1bdf1a2a6639:~$ ls -l file3.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
coder@1bdf1a2a6639:~$ diff file4.txt file3.txt
1d0
<
coder@1bdf1a2a6639:~$ diff -q file4.txt file3.txt
Files file4.txt and file3.txt differ
```

Figure 6 — `grep -i` searching a file for a string, alongside `head -1`.

8. Comparing Files

Command	Description
<code>diff file1 file2</code>	Compares two files line by line and reports the differences.
<code>diff -q f1 f2</code>	Reports only whether two files differ, without showing the details.
<code>diff -r dir1 dir2</code>	Recursively compares two directories, including their subdirectories and contents.
<code>diff -y f1 f2</code>	Displays the differences between two files side by side, in two columns.


```

coursera
File Edit Selection View Go Run Terminal Help code-server

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

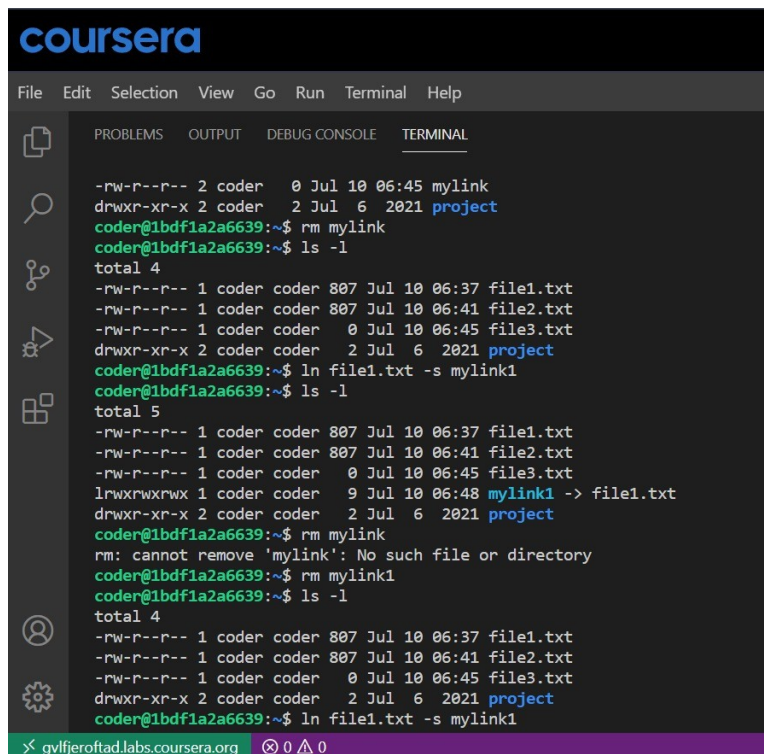
coder@1bdf1a2a6639:~$ diff -r mubashra project
diff: mubashra: No such file or directory
coder@1bdf1a2a6639:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 1 Jul 10 06:56 file4.txt
-rw-r--r-- 1 coder coder 0 Jul 10 07:09 hamna.txt
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ mkdir mubashra
coder@1bdf1a2a6639:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
-rw-r--r-- 1 coder coder 1 Jul 10 06:56 file4.txt
-rw-r--r-- 1 coder coder 0 Jul 10 07:09 hamna.txt
drwxr-xr-x 2 coder coder 2 Jul 10 07:53 mubashra
lrwxrwxrwx 1 coder coder 9 Jul 10 06:49 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ diff -r mubashra project
diff: mubashra: No such file or directory
coder@1bdf1a2a6639:~$ diff -y file3.txt file4.txt
>
coder@1bdf1a2a6639:~$ diff -y ls -l file3.txt ls -l file4.txt
diff: extra operand 'ls'
diff: Try 'diff --help' for more information.
coder@1bdf1a2a6639:~$ ls -R mubashra
mubashra:
coder@1bdf1a2a6639:~$ ls -R project
project:

```

Figure 7 — `diff -r` comparing a missing directory against `project`.

9. Links

Command	Description
ln file link	Creates a hard link: a second name for a file that shares the same inode. Deleting one name does not remove the underlying data.
ln -s file link	Creates a symbolic (soft) link: a separate file that points to the target path. Deleting the original file breaks the link.

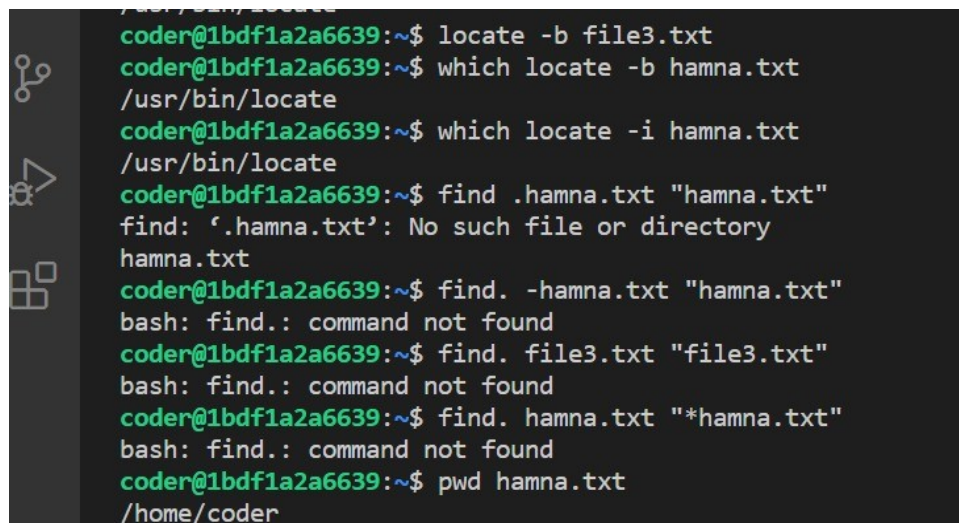


```
coursera
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
-rw-r--r-- 2 coder 0 Jul 10 06:45 mylink
drwxr-xr-x 2 coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ rm mylink
coder@1bdf1a2a6639:~$ ls -l
total 4
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ln file1.txt -s mylink1
coder@1bdf1a2a6639:~$ ls -l
total 5
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
lrwxrwxrwx 1 coder coder 9 Jul 10 06:48 mylink1 -> file1.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ rm mylink
rm: cannot remove 'mylink': No such file or directory
coder@1bdf1a2a6639:~$ rm mylink1
coder@1bdf1a2a6639:~$ ls -l
total 4
-rw-r--r-- 1 coder coder 807 Jul 10 06:37 file1.txt
-rw-r--r-- 1 coder coder 807 Jul 10 06:41 file2.txt
-rw-r--r-- 1 coder coder 0 Jul 10 06:45 file3.txt
drwxr-xr-x 2 coder coder 2 Jul 6 2021 project
coder@1bdf1a2a6639:~$ ln file1.txt -s mylink1
< gvfjroftad.labs.coursera.org 0 0
```

Figure 8 — Removing a link (mylink) and confirming the remaining files with `ls -l`.

10. Appendix — Additional Practice Screenshots

The screenshot below documents further exploration beyond this module's core command set (`locate`, `which`, and `find`), captured during the same practice session.



```
coder@1bdf1a2a6639:~$ locate -b file3.txt
coder@1bdf1a2a6639:~$ which locate -b hamna.txt
/usr/bin/locate
coder@1bdf1a2a6639:~$ which locate -i hamna.txt
/usr/bin/locate
coder@1bdf1a2a6639:~$ find .hamna.txt "hamna.txt"
find: '.hamna.txt': No such file or directory
hamna.txt
coder@1bdf1a2a6639:~$ find. -hamna.txt "hamna.txt"
bash: find.: command not found
coder@1bdf1a2a6639:~$ find. file3.txt "file3.txt"
bash: find.: command not found
coder@1bdf1a2a6639:~$ find. hamna.txt "*hamna.txt"
bash: find.: command not found
coder@1bdf1a2a6639:~$ pwd hamna.txt
/home/coder
```

Figure 9 — Exploring `locate`, `which`, and `find` for file lookup.

11. Summary

Across this module, the practiced commands cover the full lifecycle of everyday file management on Linux: creating directories and files, removing them safely, listing and reading their contents in multiple formats,

searching within them, comparing them, and linking them together. Together, these six categories form a practical foundation for working confidently in the Linux command-line environment.